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United States Patent Application Publication 20250278144
Kind Code A1
Publication Date September 04, 2025
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MIXED REALITY DISPLAY SYSTEM AND MIXED REALITY DISPLAY TERMINAL

Abstract

In a mixed reality display system, a server and a plurality of mixed reality display terminals are connected, and virtual objects are displayed. The virtual objects include a shared virtual object for which a plurality of terminals have an operation authority and a private virtual object for which only a specific terminal has an operation authority. The server has virtual object attribute information for displaying the virtual objects in each terminal, and each terminal has a motion detecting unit that detects a motion of a user for switching between the shared virtual object and the private virtual object. When a detection result by the motion detecting unit is received from the terminal, the server updates the virtual object attribute information depending on whether the virtual object is the shared virtual object or the private virtual object, and transmits data of the virtual object after the update to each terminal.

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Family ID: **64567088**

Appl. No.: **19/211756**

Filed: **May 19, 2025**

Related U.S. Application Data

parent US continuation 17349117 20210616 parent-grant-document US 12333080 child US 19211756
parent US continuation 16990027 20200811 parent-grant-document US 11068072 child US 17349117
parent US continuation 16606797 20191021 parent-grant-document US 10775897 WO continuation PCT/JP2017/020951 20170606 child US 16990027

Publication Classification

Int. Cl.: **G06F3/01** (20060101); **G02B27/01** (20060101); **G06T19/00** (20110101); **G06V20/20** (20220101)

U.S. Cl.:

CPC **G06F3/017** (20130101); **G02B27/017** (20130101); **G06T19/006** (20130101); **G06V20/20** (20220101);

Background/Summary

TECHNICAL FIELD
[0001] The present invention relates to a mixed reality display system and a mixed reality display terminal which are capable of generating and displaying a mixed reality space in which a virtually generated video, audio, or the like is combined with a video, audio, or the like of the real world and enabling a user to operate a virtually generated object in the mixed reality space.
BACKGROUND ART
[0002] As a background art in this technical field, a system that displays a shared virtual object and a personal virtual object in a mixed reality environment is disclosed in Patent Document 1. A plurality of users can perform a cooperative interaction on the shared virtual object, and the personal virtual object is visible to a single user. Further, it is described that it is possible to facilitate a cooperative interaction on the shared virtual object by a plurality of persons as the personal virtual object is provided.
CITATION LIST
Patent Document
[0003] Patent Document 1: JP 2016-525741 W
SUMMARY OF THE INVENTION
Problems to be Solved by the Invention
[0004] Specific usages of the shared virtual objects and the personal virtual objects are as follows.
[0005] A first example is a case in which a model of a prototype product or the like is displayed as the shared virtual object in the mixed reality space, and a design of the model is examined. As the virtual object is shared with a plurality of designers, discussions by a plurality of designers can be performed smoothly. On the other hand, for the shared virtual objects, an authority to change the design is given to all the designers, and thus it is not suitable for a case in which individual designers desire to examine modifications individually. In such a case, it is convenient if it is possible to switch it to the private virtual object for which a specific designer has a modification authority. To do this smoothly, it is desirable to be able to perform an operation of copying the displayed shared virtual object as the private virtual object for a specific user and an operation of merging the private virtual object for a specific user with the shared virtual object conversely.
[0006] A second example is a case in which an advertisement is displayed as the shared virtual object as in a virtual advertising catalog place. In this case, an experience as if advertisements are displayed in the same place is provided to a plurality of users in the mixed reality space. At this time, there occurs a request that a certain user desires to copy the advertisement as a personal one as can be inferred from a user action in an actual advertising catalog space. Even in such a case, it is desirable to be able to switch the shared virtual object to the private virtual object for the user. That is, the operation of switching the shared virtual object and the private virtual object (which here means a copy operation and a merge operation) is necessary.

[0007] However, in the invention disclosed in Patent Document 1, switching the shared state (the shared virtual object) and the private state (the private virtual object) alternately for a certain virtual object as described above is not taken into consideration. Further, even when switching between the shared virtual object and the private virtual object is performed, there are following problems in order to do this smoothly.
[0008] (1) It is necessary for each user to remember whether the virtual object is the shared virtual object or the private virtual object and use them differently, and users who do not remember them are likely to be confused.
[0009] (2) If the private virtual object is displayed only to a specific user A and is invisible to other user B (for example, in the case of Patent Document 1), the operation on the invisible private virtual object is not recognized by the other user B, and thus the user B feels confused since the user B does not know an operation performed by the user A.
[0010] (3) A simple means that enables the user to easily perform switching between the shared virtual object and the private virtual object (the copy operations and the merge operations) is necessary.
[0011] (4) In a case in which the user A is operating the shared virtual object using the private virtual object, operation content may be hidden by the hand of the user A and may be invisible to the other user B.
[0012] It is an object of the present invention to provide a mixed reality display system and a mixed reality display terminal which are capable of enabling the user to smoothly perform switching between the shared virtual object and the private virtual object in the mixed reality space without confusion.
Solutions to Problems
[0013] The present invention provides a mixed reality display system including a server and a plurality of mixed reality display terminals used by a plurality of users, in which the server and the plurality of mixed reality display terminals are connected via a network, virtual objects are displayed in a real space in a mixed manner, and the virtual objects include a shared virtual object for which a plurality of mixed reality display terminals have an operation authority and a private virtual object for which only a specific mixed reality display terminal has an operation authority. The server has virtual object attribute information for displaying the virtual objects in the plurality of mixed reality display terminals, and each of the plurality of mixed reality display terminals includes a motion detecting unit that detects a motion of each of the users for switching between the shared virtual object and the private virtual object. When a detection result by the motion detecting unit is received from the mixed reality display terminal, the server updates the virtual object attribute information depending on whether the virtual object is the shared virtual object or the private virtual object, and transmits data of the virtual object after the update to the plurality of mixed reality display terminals.
Effects of the Invention
[0014] According to the present invention, the user can smoothly perform switching between the shared virtual object and the private virtual object in the mixed reality space without confusion.

Description

BRIEF DESCRIPTION OF THE DRAWINGS
[0015] FIG. **1**A is a block diagram illustrating an overall configuration of a mixed reality display system according to a first embodiment.
[0016] FIG. **1**B is a diagram illustrating an example of a mixed reality space in which a shared virtual object and a private virtual object are mixed.
[0017] FIG. **2**A is a diagram illustrating a hardware configuration of mixed reality display

terminals **3***a* and **3***b*.
[0018] FIG. **2**B is a diagram illustrating a software configuration of mixed reality display terminals **3***a* and **3***b*.
[0019] FIG. **3**A is a diagram illustrating a hardware configuration of a server **1**.
[0020] FIG. **3**B is a diagram illustrating a software configuration of a server **1**.
[0021] FIG. **4**A is a data table illustrating an example of virtual object attribute information **239**.
[0022] FIG. **4**B is a data table illustrating an example of user authentication information **240**.
[0023] FIG. **4**C is a data table illustrating an example of device management information **241**.
[0024] FIG. **5** is a diagram illustrating an operation sequence in the first embodiment.
[0025] FIG. **6**A is a diagram illustrating a virtual object personalization motion (a diagram viewed from a user who is operating).
[0026] FIG. **6**B is a diagram illustrating a virtual object personalization motion (a diagram viewed from another user).
[0027] FIG. **7**A is a diagram illustrating a display state of a virtual object in a personalization motion (a diagram viewed from a user who is operating and a display pattern P**1**).
[0028] FIG. **7**B is a diagram illustrating a display state of a virtual object in a personalization motion (a diagram viewed from user who is operating and a display pattern P**2**).
[0029] FIG. **7**C is a diagram illustrating a display state of a virtual object in a personalization motion (a diagram viewed from another user and a display pattern P**3**).
[0030] FIG. **7**D is a diagram illustrating a display state of a virtual object in a personalization motion (a diagram viewed from another user and a display pattern P**4**).
[0031] FIG. **8** is a diagram describing a difference in a display position of a private virtual object (display patterns P**3** and P**4**).
[0032] FIG. **9**A is a diagram illustrating a virtual object sharing motion (a diagram viewed from a user who is operating).
[0033] FIG. **9**B is a diagram illustrating a virtual object sharing motion (a diagram viewed from another user).
[0034] FIG. **10**A is a diagram illustrating a display state of a virtual object in a sharing motion (a diagram viewed from a user who is operating and a display pattern P**1**).
[0035] FIG. **10**B is a diagram illustrating a display state of a virtual object in a sharing motion (a diagram viewed from another user and a display pattern P**3**).
[0036] FIG. **10**C is a diagram illustrating a display state of a virtual object in a sharing motion (a diagram viewed from another user and a display pattern P**4**).
[0037] FIG. **11** is a diagram describing a virtual object shape decision in a sharing motion.
[0038] FIG. **12** is a block diagram illustrating an overall configuration of a mixed reality display system according to a second embodiment.
[0039] FIG. **13**A is a diagram illustrating a hardware configuration of a mixed reality display terminal (with a built-in server function) **3***c*.
[0040] FIG. **13**B is a diagram illustrating a software configuration of a mixed reality display terminal (with a built-in server function) **3***c*.
[0041] FIG. **14** is a diagram illustrating an operation sequence in the second embodiment.
MODE FOR CARRYING OUT THE INVENTION
[0042] Hereinafter, exemplary embodiments of the present invention will be described with reference to the appended drawings.
First Embodiment
[0043] FIG. **1**A is a block diagram illustrating an overall configuration of a mixed reality display system according to a first embodiment. A mixed reality display system **10** of the present embodiment is configured such that a server **1** and a plurality of mixed reality display terminals **3***a* and **3***b* for users are connected to a network **2**. Here, there are two mixed reality display terminals, but of course, the number of mixed reality display terminals may be two or more. As the mixed

reality display terminal, for example, there is a head mount display (HMD) worn on the user's head. Hereinafter, the mixed reality display terminal is referred to simply as a “terminal.”
[0044] The server **1** has a function of holding attribute information of virtual objects, a function of performing device authentication, and a function of performing user authentication. The server **1** further includes a communication I/F and has a function n of communicating with the mixed reality display terminals (terminals) **3***a* and **3***b* via the network **2**. The server **1** has a video/audio output function and has a function of outputting/receiving the attribute information of the virtual object held by the server to/from the terminals **3***a* and **3***b* and can modify programs and functions inside the server.
[0045] On the other hand, each of the mixed reality display terminals (terminals) **3***a* and **3***b* has a communication I/F, and receives user authentication from the server **1** via the network **2** and further receives data of the virtual object and the user authentication information held in the server **1**. Each of the terminals **3***a* and **3***b* has a video output unit and an audio output unit, and has a function of outputting and displaying a video and audio of the virtual object on the basis of the data of the virtual object received from the server **1** and further modifying them.
[0046] At this time, each of the terminals **3***a* and **3***b* has a function of appropriately adjusting and outputting the video and audio so that the user has perception as if a virtual object or a virtual sound exists at a specific position in a real space when the user perceives the output video and audio. To this end, each of the terminals **3***a* and **3***b* has a means for detecting a position or acceleration information and has a function of dynamically generating a video and audio using the position/acceleration information.
[0047] FIG. **1**B is a diagram illustrating an example of a mixed reality space in which the shared virtual object and the private virtual object provided by the mixed reality display system **10** of FIG. **1** are mixed. Here, two users **4***a* and **4***b* wear the terminals (HMD) **3***a* and **3***b* on their heads, respectively, and operate the virtual object in the mixed reality space. A shared virtual object **7** and a private virtual object **8** appear in a field of view **5***a* of the user **4***a* through the terminal **3***a*. The shared virtual object **7** and the private virtual object **8** also appear in a field of view **5***b* of the user **4***b* through the terminal **3***b*. Further, the user **4***a* is performing a certain operation on the private virtual object **8**.
[0048] Next, internal configurations of the mixed reality display terminals **3***a*, **3***b* and the server **1** will be described.
[Internal Configuration of Mixed Reality Display Terminal]
[0049] FIG. **2**A is a diagram illustrating a hardware configuration of the mixed reality display terminals **3***a* and **3***b*. Here, the terminal **3***a* and the terminal **3***b* are described as having the same configuration, but they need not necessarily have the same configuration.
[0050] A distance sensor **200** detects a motion of a hand of the user or a distance to an object in the real space. A gyro sensor **201** and an acceleration sensor **202** accurately measure a position and acceleration of the terminal. A communication I/F **203** communicates with the server **1** via the network **2**. Further, an in-camera **204** detects a line of sight direction of the user, and an out-camera **205** photographs the real space.
[0051] A control unit **209** reads various types of data **212** and a program **213** held in a storage **211** through the various types of program function units **214** installed in a memory **210**, integrates various types of sensors or camera videos, and generates a video and audio of the virtual object. Then, the control unit **209** adjusts the video and audio of the virtual object so that they exist at a specific position in a specific direction in the real space.
[0052] The generated video and audio of the virtual object are output and displayed from a video output unit **206** and an audio output unit **207**. An input unit **208** receives an operation input by touch or voice. The respective blocks are connected by a bus **215**.
[0053] As a result, the user who watched the video of the terminal **3***a* or **3***b* or heard the audio can get the perception as if a virtual object or a virtual sound exists at a specific position in the real

space. Further, for a technique (mixed reality (MR)) of generating the video and audio of the virtual object so that they exist at a specific position in a specific direction in the real space, many related arts have been announced, and an example thereof is disclosed in Patent Document 1.

[0054] FIG. **2**B is a diagram illustrating a software configuration of the mixed reality display terminals **3***a* and **3***b*. Here, configurations of the memory **210** and the storage **211** in FIG. **2**A are illustrated.

[0055] The storage **211** stores various types of data **302**, a line of sight direction detection program **303**, a face position/direction detection program **304**, a video generation program **305**, an audio generation program **306**, and a communication control program **307**. These data and program are called to the memory by a device control unit **301** installed in the memory **210**, and realize various types of functions. These data and program may be stored in the memory **210** in advance.

[0056] In the following description, for the sake of simplicity, various functions realized by executing the respective programs through the device control unit **301** will be described as being realized by various types of program function units.

[0057] The line of sight direction detection program **303** detects the line of sight direction of the user on the basis of image information obtained from the in-camera **204**. Hereinafter, the line of sight direction detection program **303** and the in-camera **204** are also referred to as a "line of sight detecting unit."

[0058] A motion detection program **308** detects a motion performed by the user in the real space on the basis of information obtained from the distance sensor **200**, the out-camera **205**, or Here, examples of the motion include a motion of the like. moving a hand from a specific position to a specific position, a motion of closing a hand, and a motion of opening a hand. For the method of detecting the motion of the user from distance sensor information or camera information, many methods have been known from the past, and an example thereof is disclosed in Reference 1.

[Reference 1]

[0059] Zhou Ren, et al. "Robust hand gesture recognition with kinect sensor." Proceedings of the 19th ACM international conference on Multimedia. ACM, 2011.

[0060] Further, the motion detection program **308** may have a function for detecting a "voice motion" from a volume of a voice or speech content input to the input unit **208**. A speech recognition technique can be used as a method for detecting speech content from a voice. For the speech recognition technique, many methods have been known from the past, and an example thereof is disclosed in Reference 2.

[Reference 2]

[0061] Lawrence R. Rabiner and Biing-Hwang Juang. "Fundamentals of speech recognition." (1993).

[0062] Hereinafter, the distance sensor **200**, the out-camera **205**, the input unit **208**, and the motion detection program **308** are also referred to as a "motion detecting unit."

[0063] The face position/direction detection program **304** detects a position and direction of a face of the user in the space on the basis of signals from the gyro sensor **201** and the acceleration sensor **202**. The information is used when the virtual object and the virtual sound are generated in the video generation program **305** and the audio generation program **306**.

[0064] The video generation program **305** generates a video to be output from the video output unit **206** in accordance with information of the position or direction of the face obtained by the face position/direction detection program **304** and the image obtained from the in-camera **204**. This video is adjusted so that the user perceives it as if a virtual object exists at a specific position in the real space.

[0065] The audio generation program **306** generates an audio to be output from the audio output unit **207** in accordance with information of the position or direction of the face obtained by the face position/direction detection program **304** and the image obtained from the in-camera **204**. The audio is adjusted so that the user perceives it as if a virtual object exists at a specific position in the

real space.
[0066] The communication control program **307** communicates with the server **1** through the communication I/F **203**. Here, examples of communicated content include the attribute information (data) of the virtual object and the user authentication information, but other information may be included.
[Internal Configuration of Server]
[0067] FIG. **3**A illustrates a hardware configuration of the server **1**.
[0068] A storage **235** stores various types of data **237** for operating the server, a program **238**, and virtual object attribute information **239**, user authentication information **240**, and device management information **241** which are used for controlling the terminals **3***a* and **3***b*. The data and a program group are read out to a memory via various types of program function units **236** installed in a memory **234** and executed by a control unit **233**.
[0069] A communication I/F **231** communicates with the terminals **3***a* and **3***b* via the network **2**. Here, examples of communicated content include data of the virtual object attribute information **239** or the user authentication information **240**, but other data may be included.
[0070] An input unit **232** can receive an operation input of the user and also modify the internal data of the server **1** in accordance with a user input. A video output unit **242** and an audio output unit **243** display the internal data of the server **1** (the virtual object attribute information **239** or the like) for the user. The respective blocks are connected by a bus **244**.
[Software Configuration of Server]
[0071] FIG. **3**B illustrates a software configuration of the server **1**. Here, configurations of the memory **234** and the storage **235** in FIG. **3**A are illustrated.
[0072] The storage **235** stores various types of data **335**, a user position/direction detection program **336**, a communication control program **337**, the virtual object attribute information **239**, the user authentication information **240**, and the device management information **241**. Further, the memory **234** stores a device control unit **331**, a virtual object managing unit **332**, a user managing unit **333**, and a device managing unit **334**.
[0073] The virtual object managing unit **332** manages a position, a direction, and a shape of the virtual object and an operation authority of the user in the mixed reality space on the basis of the virtual object attribute information **239** stored in the storage **235**. The virtual object managing unit **332** modifies the virtual object attribute information **239** on the basis of the information from each of the terminals **3***a* and **3***b* obtained by the communication I/F **231**. The virtual object managing unit **332** can also modify the virtual object attribute information **239** on the basis of the operation of the user from the input unit **232**. Further, the virtual object managing unit **332** can cause information described in the virtual object attribute information **239** to be displayed for the user from the video output unit **242** or the audio output unit **243**.
[0074] The user managing unit **333** manages the user authentication information **240** for authenticating each user and the device management information **241** for authenticating a terminal owned by the user. These pieces of information **240** and **241** are stored in the storage **235**. The user managing unit **333** acquires the user authentication information such as a user name and a password from the user input in each of the terminals **3***a* and **3***b* (the input unit **208**) through the communication I/F **231** and checks whether or not it is a legitimate registered user. If the user authentication information is not correct, error information is transmitted. Further, the user name and the password may be stored in the storage **211** of each of the terminals **3***a* and **3***b* in advance, and the authentication can be performed although the user does not input them each time.
[0075] The device managing unit **334** transmits information necessary for communication with the terminals **3***a* and **3***b* to the communication control program **337** in accordance with the device management information **241** stored in the storage **235**.
[0076] Each piece of data or program stored in the storage **235** is read out to the memory **234** and executed in accordance with the control of the device control unit **331** stored in the memory **234**.

Each piece of data or program stored in the storage **235** may be stored in the memory **234** in advance.

[0077] In the following description, for the sake of simplicity, various functions realized by executing each program through the device control unit **331** will be described as being realized by various types of program function units.

[0078] The user position/direction detection program **336** detects the position and direction of each of the terminals **3***a* and **3***b* on the basis of the information obtained from each of the terminals **3***a* and **3***b* through the communication I/F **231**. These pieces of information may be transmitted to each of the terminals **3***a* and **3***b*.

[0079] The communication control program **337** controls communication with the terminals **3***a* and **3***b*. Here, examples of communicated content include the virtual object attribute information **239** and the user authentication information **240**, but other information may be included.

[Data Format]

[0080] Data formats of various types of information stored on the server **1** are described below.

[0081] FIG. **4**A is a data table illustrating an example of the virtual object attribute information **239** managed by the virtual object managing unit **332** of the server **1**. A virtual object ID **401**, a virtual object name **402**, a virtual object direction **403**, a virtual object position **404**, a virtual object color/shape **405**, a copy source **406** and a copy destination **407** indicating a copy relation between the virtual objects, and an operation authority **408** of the user are described in the virtual object attribute information **239**. A shared/private field **409** is for explanation.

[0082] The virtual object name **402**, the virtual object direction **403**, the virtual object position **404**, and the virtual object color/shape **405** are data for displaying the virtual object in the terminals **3***a* and **3***b*. Further, in this example, a virtual object ID=2, 3 is a “shared” virtual object for which all the users or a specific group has the operation authority **408**. A virtual object ID=1, 4 is a “private” virtual object for which a specific user **1** has the operation authority **408**. Further, the private virtual object ID=4 indicates a copy from the shared virtual object ID=3.

[0083] FIG. **4**B is a data table illustrating an example of the user authentication information **240** managed by the user managing unit **333** of the server **1**. A user ID (management number) **411**, a user password **412**, and the like are described in the user authentication information **240**. Further, a user name, a full name, an address, and contact information may be described. In addition to the user authentication using the user password **412**, general biometric authentication such as a finger vein, a fingerprint, a voice, a facial image, and an iris may be used, and in this case, the item of the user authentication information **240** is changed appropriately.

[0084] FIG. **4**C is a data table illustrating an example of the device management information **241** managed by the device managing unit **334** of the server **1**. A device ID (management number) **421**, a device name **422**, a device password **423**, a device IP address **424**, and the like of each terminal are described in the device management information **241**.

[Operation Sequence]

[0085] FIG. **5** is a diagram illustrating an operation sequence in the first embodiment. Here, communication between the server **1** and the terminal **3***a* (the user **4***a*) and the terminal **3***b* (the user when performing “personalization motion” and “sharing **4***b*) motion” for the virtual object will be described. Here, the personalization motion is a motion of generating the private virtual object by copying the shared virtual object and performing switching to a virtual object for which only a specific user has the operation authority. On the other hand, the “sharing motion” is a motion of merging the private virtual object with the shared virtual object and performing switching to a new shared virtual object for which each user has the operation authority.

[0086] If the server **1**, the mixed reality display terminal **3***a*, and the mixed reality display terminal **3***b* are activated, virtual object data according to the virtual object attribute information **239** is generated in the server **1** (sequence S**101**). If the user authentication information (or the device management information) is transmitted from the terminal **3***a* (the user **4***a*) to the server **1**, it is

compared with the user authentication information **240** (or the device management information **241**) in the server **1**. If the authentication is successful, the virtual object data is transmitted to the terminal **3**_a_. The terminal **3**_a_ generates the video and audio of the virtual object are through the video generation program **305** and the audio generation program **306** on the basis of the received virtual object data, and output and displayed the video and audio from the video output unit **206** and the audio output unit **207** (sequence S**102**).
[0087] On the other hand, the user authentication is similarly performed on the terminal **3**_b_ (the user **4**_b_), and if the authentication is successful, the video and audio of the virtual object are also output and displayed in the terminal **3**_b_ (sequence S**103**). Here, the displayed virtual object is the shared virtual object for which both the user **4**_a_ and the user **4**_b_ have the operation authority.
[0088] Then, if the user **4**_a_ performs, for example, an action of pulling the virtual object closer to the user in a state in which the virtual object is displayed, for example, the motion detecting unit (the sensor such as the out-camera **205** and the motion detection program **308**) of the terminal **3**_a_ detects the action as the personalization motion (sequence S**104**). Then, the gives a notification terminal **3**_a_ indicating that the personalization motion is detected to the server **1**. The server **1** updates the virtual object attribute information **239** in accordance with the personalization motion, and copies the shared virtual object to generate data of a new private virtual object (sequence $**105**). Here, the operation authority for the generated private virtual object is given only to the user **4**_a_ (the terminal **3**_a_).
[0089] The data of the new private virtual object is transmitted to the terminal **3**_a_ and the terminal **3**_b_, and each of the terminals **3**_a_ and **3**_b_ modifies the display state of the currently displayed virtual object on the basis of the received virtual object data (sequences S**106** and S**107**). As will be described later, the terminal **3**_a_ and the terminal **3**_b_ are different in the display state of the virtual object, which are managed in accordance with the virtual object attribute information **239** of the server **1**.
[0090] Further, if the user **4**_a_ performs an action of bringing the private virtual object closer to the shared virtual object in a state in which the virtual object is displayed, the motion detection unit of the terminal **3**_a_ detects the motion as the sharing motion (sequence S**108**). Then, the terminal **3**_a_ gives a notification indicating that the sharing motion is detected to the server **1**. The server **1** updates the virtual object attribute information **239** in accordance with the sharing motion, and merges the private virtual object with the shared virtual object to generate data of a new shared virtual object (sequence S**109**). Here, the operation authority for the generated shared virtual object is given to both the user **4**_a_ and the user **4**_b_.
[0091] The data of the new shared virtual object is transmitted to the terminal **3**_a_ and the terminal **3**_b_, and each of the terminals **3**_a_ and **3**_b_ modifies the display state of the currently displayed virtual object on the basis of the received virtual object data (sequences S**110** and S**111**).
[Personalization of Shared Virtual Object]
[0092] Next, the behavior when the personalization motion (S**104** in FIG. **5**) is performed will be described with reference to FIGS. **6**, **7**, and **8**.
[0093] FIGS. **6**A and **6**B are diagrams illustrating a personalization motion for switching a shared virtual object **501** to a private virtual object **502** in the mixed reality space. Here, an example in which the user **4**_a_ copies the private virtual object **502** from the shared virtual object **501** by performing the personalization motion on the shared virtual object **501** is illustrated. The personalization motion in the present embodiment is a motion of generating the private virtual object **502** for which only the user **4**_a_ has the operation authority while leaving the shared virtual object **501** as it is.
[0094] For example, when the user performs the following specific operations, the motion detecting unit (the distance sensor **200**, the out-camera **205**, and the input unit **208**) of each of the mixed reality display terminals **3**_a_ and **3**_b_ determines them as the personalization motion: [0095] A motion of pulling the shared virtual object closer while forming a specific shape with fingers is

performed. As an example, a motion of pulling the shared virtual object closer in a state in which the index finger and the thumb are connected to each other is performed. [0096] A motion of pulling the shared virtual object closer after performing a specific hand gesture is performed. As an example, a motion of pulling the shared virtual object closer after moving the hand around the circumference of the shared virtual object is performed. [0097] A motion of pulling the shared virtual object after speaking "copy" is performed. [0098] A motion of speaking "copy" after pulling the shared virtual object closer is performed. [0099] A motion of speaking "copy" during a motion of pulling the shared virtual object closer.

[0100] A mechanism in which the user can register a specific motion when performing the personalization motion may be prepared. Further, information indicating whether or not the shared virtual object **501** is copied to the private virtual object **502** by the above motion (whether or not copying is permitted) may be described in the virtual object attribute information **239**. Then, it may be controlled whether or not the virtual object is personalized in accordance with the copy permission information.

[0101] FIG. **6**A illustrates a mixed reality space **5***a* which the user **4***a* who is performing the personalization operation can see through the terminal **3***a*. On the other hand, FIG. **6**B illustrates a mixed reality space **5***b* which the other user **4***b* can see through the terminal **3***b*. As illustrated in FIGS. **6**A and **6**B, the shared virtual object **501** and the private virtual object **502** copied therefrom can be seen by all of the users **4***a* and **4***b*, but they look different to the user **4***a* and the user **4***b* (for example, a color, a shape, a distance, a voice, or the like). Accordingly, the users **4***a* and **4***b* can easily identify the shared virtual object **501** and the private virtual object **502**. Specific display states when transition from the shared virtual object **501** to the private virtual object **502** is performed will be described below with reference to FIGS. **7**A to **7**D.

[0102] FIGS. **7**A and **7**B illustrate the display state of the virtual object viewed from the user **4***a* who is performing an operation when transition from the shared virtual object **501** to the private virtual object **502** is performed. Here, a virtual sound **503** is output from the shared virtual object **501**. In FIGS. **7**A and **7**B, the display state transitions in the order of timings (1), (2), and (3). Arrows in FIGS. **7**A and **7**B indicate a motion of the hands of the user (a white arrow) and movement of the display position of the virtual object (a gray arrow).

[0103] FIG. **7**A illustrates a case according to a <display pattern P**1**>. In this case, the display position of the shared virtual object **501** is fixed to an original position during personalization transition. At the timing (1), the user **4***a* puts the hands on the shared virtual object **501**. At the timing (2), if the user **4***a* performs an action of pulling the shared virtual object **501** closer to the user **4***a*, the motion detecting unit of the terminal **3***a* determines the action as the personalization motion and transmits it to the server **1**. The server **1** copies the shared virtual object **501** to generate a new private virtual object **502** and transmits it to the terminal **3***a*. The terminal **3***a* displays the private virtual object **502**, but the display position thereof is adjusted in accordance with the position of the hand detected by the motion detecting unit. Finally, at the timing (3), the private virtual object **502** is displayed so as to be drawn to the user **4***a*.

[0104] In the process from the timings (1) to (3), the color of the shared virtual object **501** gradually fades, and the volume of the virtual sound **503** decreases. On the other hand, the newly generated private virtual object **502** is adjusted so that the color gradually increases, and the volume of the virtual sound **503** also increases. Accordingly, the user **4***a* can easily switch the shared virtual object **501** to the private virtual object **502** and easily identify the shared virtual object **501** and the private virtual object **502** generated therefrom.

[0105] FIG. **7**B illustrates a case according to a <display pattern P**2**>. In this case, the display position of the shared virtual object **501** is moved in accordance with the position of the hand of the user **4***a* and the distance of the shared virtual object **501**. A difference with the <display pattern P**1**> of FIG. **7**A will be described.

[0106] At a timing (2), if the user **4***a* performs the personalization motion, the terminal **3***a* displays the private virtual object **502** generated by the server **1** in accordance with the position of the hand of the user **4**_a_. At this time, the shared virtual object **501** is moved to a position between the original position of the shared virtual object **501** and the position of the hand and displayed (that is, moved from the original position to the position of the hand).

[0107] At a timing (3), if the position of the hand is apart from the original position of the shared virtual object **501** by a certain distance or more, the shared virtual object **501** is returned to the original position and displayed. The private virtual object **502** is displayed as if it were pulled to the user **4**_a_ as in FIG. **7**A.

[0108] FIGS. **7**C and **7**D illustrate the display state viewed from the other user **4**_b_ other than the user **4**_a_ who is performing an operation when transition from the shared virtual object **501** to the private virtual object **502** is performed. Timings (1), (2), and (3) in FIGS. **7**C and **7**D correspond to the timings (1), (2), and (3) in FIGS. **7**A and **7**B, respectively.

[0109] FIG. **7**C illustrates a case according to a <display pattern P**3**>. In this case, the position of the generated private virtual object **502** visible to the other user **4**_b_ is the same as the position visible to the user **4**_a_. In the mixed reality space **5**_b_ that is visible to the user **4**_b_, the private virtual object **502** generated by the user **4**_a_ in timing (2) (3) is displayed semi-transparently, and virtual sound is not output. In timing (3), if the distance from the shared virtual object **501** to the private virtual object **502** is d**3**, it is equal to the position seen by the user **4**_a_ (distance d**1** in FIG. **7**A or FIG. **7**B).

[0110] On the other hand, the shared virtual object **501** is normally displayed at the timings (1) and (2), but it is displayed, for example, in a darker color than a normal color, unlike the normal display, at the stage of the timing (3) after the private virtual object **502** is generated by the user **4**_a_. Accordingly, the other user **4**_b_ can easily recognize that the private virtual object **502** is copied from the shared virtual object **501** without confusion.

[0111] Further, changing the color of the shared virtual object **501** at the timing (3) is just an example, and the present invention is not limited thereto. Further, various display states are possible, for example, a contour line may be displayed in red, blinking may be performed, a mark indicating a copy may be displayed, or luminance may be changed.

[0112] FIG. **7**D illustrates a case according to a <display pattern P**4**>. In this case, the position of the generated private virtual object **502** visible to the other user **4**_b_ is different from the position visible to the user **4**_a_. That is, at the timing (3), the private virtual object **502** generated by the user **4**_a_ is displayed at a position away from the position of the hand of the user **4**_a_. At this time, the distance from the shared virtual object **501** to the private virtual object **502** is assumed to be d**4**. Accordingly, the user **4**_b_ has perception as if the private virtual object **502** were in the air.

[0113] FIG. **8** is a diagram illustrating a difference in the display position of the private virtual object **502** between FIG. **7**C and FIG. **7**D. A horizontal axis indicates the display position of the private virtual object **502** viewed from the user **4**_a_ who is performing an operation (the distance d**1** from the shared virtual object **501**), and a vertical axis indicates the display position of the private virtual object **502** viewed from the other user **4**_b_ (distances d**3** and d**4** from the shared virtual object **501**).

[0114] In the <display pattern P**3**> of FIG. **7**C, the display position (d**3**) of the private virtual object **502** viewed from the other user **4**_b_ is equal to the display position (d**1**) of the private virtual object **502** viewed from the user **4**_a_ who is performing an operation. On the other hand, in the <display pattern P**4**> of FIG. **7**D, the display position (d**3**) of the private virtual object **502** viewed from the other user **4**_b_ is smaller than the display position (d**1**) of the private virtual object **502** viewed from the user **4**_a_ who is performing an operation.

[0115] According to the <display pattern P**4**> of FIG. **7**D, at the timing (3) at which the user **4**_a_ generates the private virtual object **502**, the user **4**_a_ perceives the private virtual object **502** which is closer to the user **4**_a_ as if it were in the air.

[0116] The advantage of the display method of the <display pattern P**4**> lies in that while the user

4*a* is operating the private virtual object **502** at a position close to the user **4***a*, the user **4***b* can observe the private virtual object **502** without being hidden by the hand of the user **4***a*. Further, it is possible to easily identify that the private virtual object **502** is copied from the shared virtual object **501** and generated without confusion.

[0117] In FIGS. **7**C and **7**D, in order to cause the other user **4***b* to surely recognize that the shared virtual object **501** is switched to the private virtual object **502**, the following methods may be used.

[0118] There are cases in which the user **4***b* sees another direction at a time at which the user **4***a* performs the personalization motion. In this case, it is desirable to detect the line of sight direction of the user **4***b* through the line of sight detecting unit (the in-camera **204** and the line of sight detection program **303**) installed in the terminal **3***b* of the user **4***b* and switch the display state (for example, the color) of the shared virtual object **501** at a timing at which the line of sight of the user **4***b* is placed on the shared virtual object **501**. Accordingly, the user **4***b* does not overlook the personalization switching of the shared virtual object **501**.

[0119] Further, when the position at which the user **4***a* performs the personalization motion is far from the position of the user **4***b*, it is desirable to detect the distance between the shared virtual object **501** and the terminal **3***b* through the distance sensor **200** and switch the display state (for example, the color) of the shared virtual object **501** at a timing at which the distance becomes a specified distance. Accordingly, the user **4***b* does not overlook the personalization switching of the shared virtual object **501**.

[0120] Also, it is desirable to detect the speed of the acceleration at which the user **4***b* (the terminal **3***b*) approaches the shared virtual object **501** through the acceleration sensor **202** and switches the display state (for example, the color) when a predetermined speed or predetermined acceleration is exceeded. Accordingly, when the user **4***b* approaches the shared virtual object **501**, the user **4***b* does not overlook that the virtual object is switched to the private virtual object.

[Sharing of Private Virtual Object]

[0121] Next, a behavior when the sharing motion detection is performed (S**108** in FIG. **5**) will be described with reference to FIGS. **9**, **10**, and **11**.

[0122] FIG. **9**A and FIG. **9**B are diagrams illustrating a sharing motion for switching a private virtual object **601** to a shared virtual object **602** in the mixed reality space. Here, an example in which the user **4***a* merges the private virtual object **601** with the shared virtual object **602** by performing the sharing motion on the private virtual object **601** is illustrated. The sharing motion in the present embodiment is a motion of generating one new shared virtual object **602** from the private virtual object **601** and the shared virtual object **602**, reflecting a feature of the private virtual object **601**, and giving the operation authority even to the other user **4***b* by sharing.

[0123] For example, when the user performs the following specific motions, the motion detecting unit of each of the mixed reality display terminals **3***a* and **3***b* determines them as the sharing motion. [0124] A motion of bringing the private virtual object closer to the shared virtual object while forming a specific shape with fingers is performed. As an example, a motion of bringing the private virtual object closer to it with the index finger and the thumb are connected to each other is performed. [0125] A motion of bringing the private virtual object closer to the shared virtual object after performing a specific hand gesture is performed. As an example, a motion of bringing the private virtual object closer to the shared virtual object after moving the hand around the circumference of the private virtual object is performed. [0126] A motion of bringing the private virtual object closer to the shared virtual object after speaking “merge” is performed. [0127] A motion of speaking “merge” after bringing the private virtual object closer to the shared virtual object is performed. [0128] A motion of speaking “merge” during a motion of bringing the private virtual object closer to the shared virtual object is performed.

[0129] A mechanism in which a user can register a specific motion when performing the sharing motion may be prepared.

[0130] FIG. **9**A illustrates the mixed reality space **5***a* which the user **4***a* who is performing the

sharing operation can see through the terminal **3**_a_. On the other hand, FIG. **9**B illustrates the mixed reality space **5**_b_ which the other user **4**_b_ can see through the terminal **3**_b_. FIGS. **9**A and **9**B illustrate the operation of merging the two virtual objects by bringing the private virtual object **601** closer to the shared virtual object **602**. Here, the private virtual object **601** and the shared virtual object **602** are displayed so that they look different to the user **4**_a_ and the user **4**_b_ (for example, a color, a shape, a distance, a voice, or the like). Accordingly, the users **4**_a_ and **4**_b_ can easily identify the private virtual object **601** and the shared virtual object **602**. Specific display states when transition from the private virtual object **601** to the shared virtual object **602** is performed will be described below with reference to FIGS. **10**A to **10**C.
[0131] FIG. **10**A illustrates a display state of the virtual object viewed from the user **4**_a_ who is performing an operation when transition from the private virtual object **601** to the shared virtual object **602** is performed. Here, a virtual sound **603** is output from the private virtual object **601**. In FIG. **10**A, the display state transitions in the order of timings (1), (2), and (3). Arrows in FIGS. **7**A and **7**B indicate a motion of the hands of the user (a white arrow) and movement of the display position of the virtual object (a gray arrow). This display pattern corresponds to the <display pattern **1**> of FIG. **7**A.
[0132] At a timing (1), the user **4**_a_ puts the hands on the private virtual object **601**. At a timing (2), if the user **4**_a_ performs an action of bringing the private virtual object **601** closer to the shared virtual object **602**, the motion detecting unit of the terminal **3**_a_ determines the action as the sharing motion and transmits it to the server **1**. The server **1** merges the private virtual object **601** and the shared virtual object **602** which is its copy source to generate a new shared virtual object **602** and transmits it to the terminal **3**_a_. At a timing (3), the terminal **3**_a_ displays the new shared virtual object **602** in the original display state (dark color).
[0133] In the process from the timings (1) to (3), the shared virtual object **602** gradually becomes darker, and the volume of the virtual sound **604** increases. On the other hand, the private virtual object **601** is adjusted so that the color gradually fades, and the volume of the virtual sound **603** is reduced. Accordingly, the user **4**_a_ can easily switch the private virtual object **601** to the shared virtual object **602**.
[0134] FIGS. **10**B and **10**C illustrate a display state of the virtual object viewed from the other user **4**_b_ when transition from the private virtual object **601** to the shared virtual object **602** is performed. Timings (1), (2), and (3) in FIGS. **10**B and **10**C correspond to the timing (1), (2), and (3) in FIG. **10**A, respectively.
[0135] FIG. **10**B illustrates a case according to a <display pattern P**3**>. In this case, as described above with reference to FIG. **8**, the position of the private virtual object **601** visible to the other user **4**_b_ is the same as the position visible to the user **4**_a_.
[0136] At a timing (1), in the mixed reality space **5**_b_ that is visible to the user **4**_b_, the private virtual object **601** of the user **4**_a_ is translucent, and the shared virtual object **602** is displayed in a dark color. That is, a distance d**3** from the shared virtual object **602** to the private virtual object **601** is equal to the position (a distance d**1** in FIG. **10**A) viewed by the user **4**_a_. If the user **4**_a_ performs a motion of bringing the private virtual object **601** closer to the shared virtual object **602** at a timing (2), both virtual objects are merged at a timing (3), and a new shared virtual object **602** is displayed in a normal color.
[0137] Accordingly, the user **4**_b_ can easily identify the private virtual object **601** and the shared virtual object **602** and easily recognize that the shared virtual object **602** is merged from the private virtual object **601** without confusion.
[0138] FIG. **10**C illustrates a case according to a <display pattern P**4**>. In this case, as described in FIG. **8**, the position of the private virtual object **601** visible to the other user **4**_b_ is different from the position visible to the user **4**_a_. That is, at a timing (1), a distance d**4** from the shared virtual object **602** to the private virtual object **601** is smaller than a position (the distance d**1** in FIG. **10**A) viewed by the user **4**_a_. Accordingly, the user **4**_b_ has perception as if the private virtual object **601** were in

the air. Therefore, the user **4***b* can observe the private virtual object **601** without being hidden by the hand of the user **4***a*.

[0139] FIG. **11** is a diagram for describing a virtual object shape decision in the sharing motion. Commonly, a shape of the private virtual object **601** is different from a shape of the shared virtual object **602** which is merged with it. Therefore, when these virtual objects are merged, the user **4***a* can sequentially check whether or not a new virtual object is employed (combined) for parts which do not include each other in a difference between the shapes of the respective virtual objects.

[0140] At a timing (1), the private virtual object **601** is copied from the shared virtual object **602**. After copying, a new private virtual object **601***a* and a private virtual object **601***b* are generated in accordance with an operation of the user **4***a* and combined with the private virtual object **601**. Here, the operation of merging an aggregate of the private virtual objects **601**, **601***a*, and **601***b* into the shared virtual object **602** of the copy source is performed.

[0141] At a timing (2), the user **4***a* selects whether or not the private virtual object **601**, which is a first difference is combined with the shared virtual object **602**. For this selection process, a method such as a hand gesture or speech recognition is used. For example, the user **4***a* is assumed to select to combine the private virtual object **601***a* with the shared virtual object **602**. At a timing (3), the user **4***a* selects whether or not the private virtual object **601***b* which is a second difference is combined with the shared virtual object **602**. Here, the user **4***a* rejects the combination of the private virtual object **601***b*. As a result, finally, as in a timing (4), a new set of shared virtual objects in which the private virtual object **601***a* is combined with the shared virtual object **602** is generated.

[0142] Further, in FIG. **11**, a case in which the newly generated private virtual object is combined has been described, but even when the private virtual object itself is modified, an operation of selecting whether or not the private virtual object is merged into the shared virtual object can be performed in a similar procedure.

[0143] As described above, according to the first embodiment, in the mixed reality space, the switching between the shared virtual object and the private virtual object can be performed smoothly without confusion including not only the user who is performing an operation but also other users.

Second Embodiment

[0144] In a second embodiment, all the functions performed by the server **1** in the first embodiment are installed in the mixed reality display terminal. Therefore, the server **1** in the first embodiment is unnecessary. A difference from the first embodiment will be described below.

[0145] FIG. **12** is a block diagram illustrating an overall configuration of a mixed reality display system according to the second embodiment. A mixed reality display system **10′** in the present embodiment includes a network **2** and a plurality of mixed reality display terminals **3***c* and **3***b* for the users. Among them, the mixed reality display terminal **3***c* is a terminal with a built-in server function.

[0146] FIG. **13**A is a diagram illustrating a hardware configuration of the mixed reality display terminal (with a built-in server function) **3***c*. The terminal **3***b* has the same configuration as that of FIG. **2**A in the first embodiment. In the terminal **3***c*, the virtual object attribute information **239**, the user authentication information **240**, and the device management information **241** installed in the server **1** are additionally stored in the storage **211**. The roles of the respective blocks are identical to those described in FIG. **3**A in the first embodiment.

[0147] FIG. **13**B is a diagram illustrating a software configuration of the mixed reality display terminal (with a built-in server function) **3***c*. The terminal **3***b* has the same configuration as that of FIG. **2**B in the first embodiment. Here, configurations of the memory **210** and the storage **211** in FIG. **13**A are illustrated.

[0148] In the terminal **3***c*, the virtual object managing unit **332**, the user managing unit **333**, and the device managing unit **334** installed in the server **1** are additionally stored in the memory **210**. Further, the user position/direction detection program **336**, the virtual object attribute information

239, the user authentication information **240**, and the device management information **241** installed in the server **1** are additionally stored in the storage **211**. The roles of the respective blocks are identical to those described in FIG. **3**B in the first embodiment.

[0149] FIG. **14** is a diagram illustrating an operation sequence in the second embodiment. In this operation sequence, the terminal **3***c* also serves as the server **1** in the first embodiment. Therefore, communication between the terminal **3***a* and the server **1** in the first embodiment (FIG. **5**) is omitted. That is, when the personalization motion (S**104**) or the sharing motion (S**108**) is performed in the terminal **3***c*, the built-in virtual object attribute information **239** is updated (S**105** and S**109**), and the display state of the virtual object displayed by itself modified (S**106** and S**110**). Also, the updated virtual object data is transmitted to the terminal **3***b*. The other operations are the same as those in the first embodiment (FIG. **5**).

[0150] Although FIG. **14** illustrates a case in which the personalization motion and the sharing motion are performed in the terminal **3***c*, when the personalization motion and the sharing motion are performed in the terminal **3***b*, the detection information of the personalization motion and the sharing motion is transmitted to the terminal **3***c*, and the virtual object attribute information **239** stored in the terminal **3***c* is updated in a similar manner as described above.

[0151] According to the second embodiment, if the mixed reality display terminal is connected to the network, a plurality of users can share the same mixed reality space and easily perform the cooperative operations on the virtual objects.

[0152] In the above embodiment, one mixed reality display terminal **3***c* plays the role of the server, but all the mixed reality display terminals may play the role of the server and share the virtual object attribute information.

[0153] The present invention is not limited to the above-described embodiments and includes various modifications. The above-described embodiments are detailed explanation of the entire system in order to facilitate understanding of the present invention, and the present invention is not necessarily limited to those having all the configurations described above. Further, addition, deletion, or substitution of other components may be performed on some components of each embodiment.

[0154] Further, some or all of the components, the functions, the processing units, the processing means, or the like described above may be realized by hardware by designing them, for example, with an integrated circuit. Further, the components, the functions, or the like described above may be realized by software by interpreting and executing a program that realizes each function through a processor. A program that realizes each function and information such as a table or a file can be stored in a memory, a recording device such as a hard disk or a solid-state drive (SSD), or a recording media such as an IC card, an SD cards, or a DVD.

[0155] Further, control lines and information lines considered necessary for explanation are illustrated, but all control lines and all information lines in a product need not be necessarily illustrated. In practice, almost all the components may be considered to be connected to one another.

REFERENCE SIGNS LIST

[0156] **1** Server [0157] **2** Network [0158] **3***a*, **3***b* Mixed reality display terminal (terminal) [0159] **3***c* Mixed reality display terminal (with built-in server function) [0160] **4***a*, **4***b* User [0161] **7**, **8**, **501**, **502**, **601**, **602** Virtual object [0162] **10**, **10**′ Mixed reality display system [0163] **200** Distance sensor [0164] **203**, **231** Communication I/F [0165] **204** In-camera [0166] **205** Out-camera [0167] **206**, **242** Video output unit [0168] **207**, **243** Audio output unit [0169] **208** Input unit [0170] **209**, **233** Control unit [0171] **210**, **234** Memory [0172] **221**, **235** Storage [0173] **239** Virtual object attribute information [0174] **332** Virtual object managing unit

Claims

1. A mixed reality display system in which virtual objects are displayed in real space in a mixed manner, wherein, in the mixed reality display system, a plurality of the mixed reality display terminals used by a plurality of users are connected to a server, the virtual objects include a shared virtual object for which an operation authority is given in accordance with an attribute information and a private virtual object for which only a specific mixed reality display terminal has an operation authority in accordance with an attribute information, the mixed reality display terminal generates a switching operation instruction by the user for switching between the shared virtual object and the private virtual object, the mixed reality display terminal transmits information to the server regarding the switching operation instruction to the attribute information of the virtual object necessary for displaying and operating the virtual object on each mixed reality display terminal, the server receives the switching operation instruction by the user, and changes the operation authority as follows: i) when switching the shared virtual object to the private virtual object, the operation authority is given only to a specific mixed reality display terminal in accordance with the attribute information, or, ii) when switching the private virtual object to the shared virtual object, the operation authority is given to each mixed reality display terminal in accordance with the attribute information, the server transmits updated attribute information to the mixed reality display terminals, and change the appearance of the virtual object in color, luminance, or shape, in order to indicate the difference between shared and private, and the mixed reality display terminal receives updated attribute information from the server based on the information about the switching operation instruction.
2. The mixed reality display system according to claim 1, wherein, in a process of switching a display based on the switching operation instruction by the user, each of the plurality of mixed reality display terminals displays the same virtual objects by gradually changing the color, the luminance, or the shape.
3. The mixed reality display system according to claim 1, wherein the mixed reality display terminal that has received the switching operation instruction of the virtual object by the user and the mixed reality display terminal that has not received the switching operation instruction of the virtual object by the user display the same virtual objects at different positions in accordance with a distance between the virtual object and the user.
4. The mixed reality display system according to claim 1, comprising a line of sight detecting unit that detects a line of sight direction of the user, and the line of sight detecting unit detects that the line of sight of the user is placed on the virtual object, and then a display state for the virtual object is switched in accordance with the virtual object attribute information.
5. The mixed reality display system according to claim 1, comprising a distance detecting unit that detects a distance from the virtual object, and the distance detecting unit detects that the mixed reality display terminal approaches the virtual object by a predetermined distance, and then the virtual object is displayed with a different color, luminance, or shape in accordance with the virtual object attribute information.
6. The mixed reality display system according to claim 1, comprising a speed detecting unit that detects a speed or acceleration approaching the virtual object, and the speed detecting unit detects that the mixed reality display terminal approaches the virtual object by a distance beyond a predetermined speed or acceleration, and then the virtual object is displayed with a different color, luminance, or shape in accordance with the virtual object attribute information.